

# ShopPulse — E-Commerce Sales Analytics Platform

Comprehensive Project Documentation, Architecture & Strategic Business Case Study

**Author:** B.Tech AI & Data Science Student | **Target Role:** Data Analyst / Analytics Engineer  
**Dataset Scope:** 12,500 Transactions | \$3.61M Revenue | \$1.59M Profit | 43.93% Margin | 20 Advanced SQL Queries

## 1. Executive Summary & Business Objectives

**ShopPulse** is an enterprise-grade omnichannel e-commerce retail platform selling across 5 core merchandise divisions (Technology, Furniture, Apparel, Office Supplies, Home & Kitchen) and 4 geographic regions (North, South, East, West). As the business expanded across 12,500+ transactions and \$3.6M+ in revenue, leadership commissioned an end-to-end data analytics system to evaluate sales velocity, customer lifetime value, promotional discount elasticity, and profit margin health.

**Core Analytics Goals:**

- **Revenue & Profit Dynamics:** Track monthly sales velocity and seasonal holiday surges.
- **Margin Optimization:** Diagnose categories driving gross cash vs. those suffering margin erosion.
- **Promotional Guardrails:** Quantify profit loss from excessive discounting (>15%).
- **Customer Segmentation:** Classify customer retention cohorts and calculate RFM / CLV metrics.
- **Inventory Rationalization:** Evaluate Pareto (80/20) catalog revenue skew.

## 2. Enterprise Performance Scorecard

| Metric Name               | Value           | Business Benchmark | Analytical Interpretation                                                     |
|---------------------------|-----------------|--------------------|-------------------------------------------------------------------------------|
| Total Revenue             | \$3,613,656.82  | +14.2% MoM         | Aggregate 18-month top-line cash flow across 12,500 transactions.             |
| Total Gross Profit        | \$1,587,554.34  | +11.8% MoM         | Net earnings generated after deduction of Cost of Goods Sold (COGS).          |
| Overall Profit Margin     | 43.93%          | Target: >40.0%     | Healthy unit economics benchmark with high margin expansion potential.        |
| Total Order Volume        | 12,500 Orders   | +8.4% Volume       | Steady transaction flow averaging 694 orders per month.                       |
| Unique Customers          | 2,108 Customers | 2.1K Accounts      | Broad customer acquisition footprint across 4 geographic regions.             |
| Average Order Value (AOV) | \$289.09        | Target: \$300.00   | Mean basket spend per completed checkout transaction.                         |
| Repeat Customer Rate      | 81.83%          | Benchmark: >75%    | Exceptional customer retention: 1,725 out of 2,108 buyers placed >= 2 orders. |

## 3. Technical Stack & Data Pipeline Architecture

The project implements a full modern data stack lifecycle from ingestion to interactive visualization:

| Layer               | Technology            | Key Responsibilities & Components                                                                       |
|---------------------|-----------------------|---------------------------------------------------------------------------------------------------------|
| Data ETL & Cleaning | Python, Pandas, NumPy | Deduplication (150 dups pruned), relational entity imputation, calendar enrichment, financial formulae. |

|                    |                                    |                                                                                                       |
|--------------------|------------------------------------|-------------------------------------------------------------------------------------------------------|
| Database Layer     | PostgreSQL 16 & SQLite 3           | Star Schema relational tables (dim_customers, dim_products, fact_orders) with B-Tree indexes and      |
| SQL Analytics      | Advanced SQL (20 Queries)          | CTEs, Window Functions (DENSE_RANK, LAG, SUM OVER), Cohorts, RFM Scoring, CLV Modeling                |
| Visualization & BI | Streamlit, Plotly, Flask, Chart.js | Dual Dashboards: Multi-view Streamlit application (port 8501) and zero-dependency local web app (p    |
| Quality Assurance  | Pytest Suite                       | 14 automated unit & integration tests validating ETL pipelines, math integrity, and query performance |

## 4. Advanced SQL Analytics Catalog (20 Production Queries)

Below is a categorized selection of the 20 production SQL queries implemented in database/analysis\_queries.sql demonstrating analytics engineering competencies:

### Query 03: Month-Over-Month (MoM) Growth Analysis using LAG()

*Business Purpose: Computes period-over-period sales acceleration and margin expansion rates without manual self-joins.*

```
WITH monthly_metrics AS (  
    SELECT SUBSTR(order_date, 1, 7) AS year_month,  
           ROUND(SUM(sales), 2) AS revenue, ROUND(SUM(profit), 2) AS profit  
    FROM fact_ecommerce_sales GROUP BY SUBSTR(order_date, 1, 7)  
)  
SELECT year_month, revenue,  
       LAG(revenue, 1) OVER (ORDER BY year_month) AS prev_month_revenue,  
       ROUND(((revenue - LAG(revenue, 1) OVER (ORDER BY year_month)) /  
              LAG(revenue, 1) OVER (ORDER BY year_month)) * 100.0, 2) AS mom_revenue_growth_pct  
FROM monthly_metrics ORDER BY year_month ASC;
```

### Query 07: Category Product Ranking using DENSE\_RANK()

*Business Purpose: Identifies top 3 revenue leaders inside each category to guide catalog merchandising.*

```
WITH category_product_sales AS (  
    SELECT category, product_name, ROUND(SUM(sales), 2) AS total_sales,  
           DENSE_RANK() OVER (PARTITION BY category ORDER BY SUM(sales) DESC) AS rank_in_cat  
    FROM fact_ecommerce_sales GROUP BY category, product_name  
)  
SELECT category, rank_in_cat, product_name, total_sales  
FROM category_product_sales WHERE rank_in_cat <= 3 ORDER BY category, rank_in_cat ASC;
```

### Query 11: Customer Lifetime Value (CLV) by Customer Segment

*Business Purpose: Evaluates long-term monetary value across Corporate, Consumer, Home Office, and Small Business accounts.*

```
WITH customer_aggregates AS (  
    SELECT customer_id, customer_segment, COUNT(DISTINCT order_id) AS total_orders,  
           SUM(sales) AS total_spend, SUM(profit) AS total_profit  
    FROM fact_ecommerce_sales GROUP BY customer_id, customer_segment  
)  
SELECT customer_segment, COUNT(customer_id) AS total_customers,  
       ROUND(AVG(total_orders), 1) AS avg_orders, ROUND(AVG(total_spend), 2) AS avg_clv_revenue,  
       ROUND(SUM(total_spend), 2) AS segment_revenue  
FROM customer_aggregates GROUP BY customer_segment ORDER BY segment_revenue DESC;
```

### Query 18: Pareto 80/20 Cumulative Product Revenue Share

*Business Purpose: Identifies high-concentration SKUs responsible for the first 80% of aggregate revenue.*

```
WITH product_totals AS (  
    SELECT product_name, category, ROUND(SUM(sales), 2) AS product_sales  
    FROM fact_ecommerce_sales GROUP BY product_name, category  
)  
ranked_products AS (  
    SELECT product_name, product_sales,  
           SUM(product_sales) OVER (ORDER BY product_sales DESC ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT  
ROW) AS running_sales,  
           SUM(product_sales) OVER () AS total_enterprise_sales  
    FROM product_totals
```

```
)  
SELECT product_name, product_sales, ROUND((running_sales / total_enterprise_sales) * 100.0, 2) AS cumulative  
_pct  
FROM ranked_products LIMIT 15;
```

## 5. Top Strategic Business Insights & Recommendations

Extracted from reports/business\_insights.md, these 6 core case studies illustrate analytical discovery to quantifiable C-Suite recommendations:

### 1. Category Margin Asymmetry

- **Finding:** Technology accounts for 49.82% (\$1.80M) of revenue with a 39.53% margin, while Apparel represents 9.74% (\$352K) with a high 62.66% margin.
- **Impact:** Heavy hardware reliance limits blended margin expansion. Shifting volume to Apparel yields higher profit per advertising dollar.
- **Recommendation:** Reallocate 15% of marketing spend toward Apparel; introduce tech-apparel bundle packs. Target

### 2. Q4 Seasonality & Inventory Risk

- **Finding:** November (\$298K) and December (\$325K) generate +74.6% higher monthly revenue than baseline months.
- **Impact:** Supply chains risk catastrophic stockouts on top 20 SKUs during peak holiday weeks.
- **Recommendation:** Finalize purchase orders with tier-1 suppliers by early September (60-day buffer); establish dynamic safety stock by Oct 15.

### 3. Deep Discount Margin Cannibalization

- **Finding:** Discounts >15% cause profit margin to plummet from 46.8% down to 24.2%, with only a 11.4% gain in order volume.
- **Impact:** Markdown pricing destroys profit margins without generating compensating unit elasticity.
- **Recommendation:** Cap promotional discounts at 12%; shift to threshold-based volume tiers (e.g. '\$25 off orders above \$200').

### 4. Customer Retention & Repeat Purchase Engine

- **Finding:** 81.83% of unique customers (1,725/2,108) are repeat buyers; accounts with 5+ orders drive 58.2% of total revenue.
- **Impact:** Customer retention economics are substantially cheaper than top-of-funnel customer acquisition (CAC).
- **Recommendation:** Launch automated ShopPulse VIP loyalty tiers and automated post-purchase email re-order triggers at Day 14, 30, and 60.

### 5. B2B / Corporate Highest Lifetime Value

- **Finding:** Corporate clients generate an AOV of \$342.10 and average annual CLV of \$2,140 with 1.4x higher order frequency.
- **Impact:** Corporate accounts represent the most predictable, recurring revenue stream with lower price sensitivity.
- **Recommendation:** Build dedicated B2B self-service purchasing portals with automated GST invoicing and assign dedicated account managers.

### 6. Pareto 80/20 SKU Catalog Concentration

- **Finding:** The top 19.8% of product SKUs (127/640) account for 78.4% of total enterprise revenue.
- **Impact:** Working capital is tied up in low-velocity long-tail inventory with high holding costs.
- **Recommendation:** Phase out bottom 10% non-performing SKUs and transition slow catalog items to drop-ship vendor models.

## 6. Interview Talking Points & Technical Elevator Pitch

How to present this project in a Data Analyst / BI Engineer interview:

*"In this project, I built **ShopPulse**, an end-to-end e-commerce analytics platform analyzing 12,500 transactions across 18 months. I engineered a data cleaning pipeline in Python to resolve duplicate orders, missing values, and enforce mathematical financial constraints. Next, I designed a normalized Star Schema in PostgreSQL/SQLite and authored 20 advanced SQL queries utilizing Window Functions (LAG, DENSE\_RANK, SUM OVER) to model Customer Lifetime Value, RFM segmentation, and MoM growth rates. Finally, I built interactive dashboards in Streamlit and delivered 12 quantified business recommendations that identified how reallocating marketing to Apparel (62.7% margin) and capping discounts at 12% directly expands enterprise profit margins."*

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**GitHub Repository:** <https://github.com/mahamoodmajin183-ui/shop-pulse-ecommerce-analytics> | **License:** MIT